ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

On Combiners for Verifiable Random Functions

Bachelor Thesis

Basil Rohner

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Advisors: Prof. Dr. A. D. Visor, Dr. P. Ostdoc
Department of Computer Science, ETH Zürich

Acknowledgments

Abstract

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Chapter 1

Introduction

1.1 Motivation and Previous Work

Verifiable Random Functions. A verifiable random function (VRF) introduced by Micali, Rabin and Vadhan in [**?**] is a cryptographic primitive which allows the holder of a secret key to evaluate a pseudorandom function at arbitrary points and obtain image-proof pairs. In the absense of the proof the images are computationally indistinguishable from random elements in the range while the corresponding proof allows the owner of the corresponding public key to verify whether the output was indeed computed correctly. Additionally we require that the verification accepts an image-proof pair if and only if the image is the unique evaluation of the point under the given secret key. As such, a VRF can be considered the public-key equivalent of the more common secret-key pseudorandom function (PRF).

In practice VRFs are used to implement signatures with strong verifiability guarantees [**?**], resettable zero-knowledge proofs [**?**], lottery and e-cash systems [**?**, **?**, **?**] and in Proof-of-Stake blockchains through distributed leader election protocols [**?**, **?**, **?**, **?**].

In addition to standard VRFs, both stronger variants such as constrainable VRFs which additionally allow the constraining of secret keys to a specified subset of evaluations [**?**], key-homomorphic and aggregable VRFs that allow the aggregation of multiple image-proof pairs into a single image-proof pair [**?**] and unbiasable VRFs that guarantee unbiasability, preventing adversaries to generated keys such that the corresponding evaluations can be biased [**?**].

We also consider weaker notions of VRFs, such as verifiable unpredictable functions (VUFs) already introduced in [**?**] together with VRFs and weakly verifiable random functions (wVRFs) which only require a weaker notion of pseudorandomness, that is pseudorandomness on uniformly random evaluation points. While VRFs can be constructed from VUFs using hardcore

bits [**?**], the authors of [**?**] show that VRFs can not, in general, be constructed from wVRFs using a seperation relying on enhanced trapdoor permutations.

Robust (Black-Box) Combiners. Robust combiners are constructions that take many instances of the same cryptographic primitive, called candidate schemes, in form of a description of their Turing machine, and securely implement that cryptographic primitive if a threshhold of the candidate schemes are secure. Since we generally intend for the combiner to derive its security from the subset of secure candidate schemes, and not by implementing a secure instance of the underlying primitive itself and ignoring the candidates, we require black-box reductions from the combiner construction to any sufficiently large subset of its candidates. In this work mainly consider robust black-box combiners, a subclass of combiners that restricts access of the combiner to its candidate schemes to oracles which can be queried but not inspected for their underlying description. Robust combiners allow the hedging of trust in cryptographic hardness assumptions underlying computational security properties, by basing their security on a subset of hard problems rather than a a single hard problem. In practice robust combiners are uesful to reduce the dependency of a systems on the security of a specific third party implementation, e.g. a cryptographic library. In theory robust combiners are related to universal cryptographic construction, i.e. constructions that implement a secure instance of a cryptographic primitive under the only assumption that a secure instance of the cryptographic primitive exists, a prominent example is Levin's universal one-way function[**?**]. Finally combiners play a central role in the transition towards post-quantum cryptography where hardness assumptions are not as well studied as the respective assumptions for the classic cryptography counterpart, in particular the combination of classical and post-quantum cryptographic assumptions has been a fruitful research field.

Combiners have been considered for a wide variety of cryptographic primitivey, including key exchange protocols [**?**], signature schemes [**?**], oblivious-transfer [**?**], zero-knowledge proofs [**?**], commitment schemes [**?**], and indistinguishability obfuscation [**?**]. While pseudorandom functions (PRFs) admit easy robust $(n;1)$ combiners by simply XOR'ing the outputs of the candidate schemes, evaluated on independent keys the combination of VRFs, the combination of VRFs is less obvious. To the best of our knowledge, there has been no prior research on robust combiners for verifiable random functions.

Our Contribution

We introduce a new definition of a robust black-box combiner, a robust black-box DAG combiner, which models the implemented algorithms of

the combiner as computations that follows the structure of a vertex-colored directed acyclic graph starting from a source and ending in a sink, where for each vertex in the graph we query a specific oracle of a specific candidate on the output of a probabilistic polynomial time Turing machine of all predecessors. We show that the notion of a robust black-box DAG combiner is sufficiently strong to capture all ppt. robust black-box combiners. We use this new model to show the impossibility of combining multiple instances of wVRFs which extends to the impossibility of combining VUFs and VRFs without additional assumptions. We argue that the combination of the unique provability property is however less important since it is of information theoretic or statistical nature and therefore doesn't rely on any computational hardness assumption. In fact, good VRF constructions such as ECVRF [**?**], lattice VRF [**?**] and [**?**] mathematically prove unique provability mathematically, rendering combination less important. We consider the possibility to combine more advanced security properties on VRFs such as unbiasability for which we construct a combiner for uniform constructions which relies on collision resistant hash functions and randomness extractors to combine unbiasable VRFs under the additional assumption that all candidate schemnes are uniquely provable. We also investigate combiners for secret-key aggregability and public-key aggregability. Secret-key aggregability is a publicly verifiable property due to an efficient algorithm by Blum, Luby and Rubinfeld [**?**] which allows checks for the homomorphism property of a function in time independent of the size of the underlying groups. Combiners for publiy-key aggregability Finally we investigate additional properties such as constrainability [**?**] nd which all admit the simple pseudorandomness combiner construction.

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1.2 Technical Overview

Robust Black-Box DAG Combiner. We present another specification of robust black-box combiners, termed robust black-box DAG combiners, which describe a form of black-box combiners that are structured by a directed acyclic graph for each implemented algorithm of the underlying cryptographic primitive in **??**. We show that the newly introduced notion in **??** is in fact equivalent to the standard definition of robust black-box combiners **??**, a result we formalize in the following theorem.

restate

Theorem 1.1

Impossibility of Black-Box VRF Combiner. Subsequently we show that there exists no robust $(n;k)$ black-box combiner that takes as input n candidates, of which k securely implement a VRF according to **??** the combiner implements a VF in **??** that is statistically correct according to **??**, uniquely provable according to **??**, and weakly pseudorandom according to **??**. That is

even assuming that $n-1$ candidates implement secure VRFs according to **??** the combiner cannot guarantee to implement a statistically correct weak VRF. We formally state this in the following theorem. restate

Theorem 1.2

Relaxed Combiners for VRF. Having established **??** we move on by assuming that every candidate scheme is uniquely provable, which enables a simple robust $(n;1)$ black-box combiner for VRF in **??** which samples independent keys per candidate generically for the key-generation algorithm, for evaluation the combiner evaluates every candidate on the input with their corresponding secret key, returns the XOR of the images with the proof contiaining all views of the candidate schemes. For verification the combiner verifies that all candidates views in the proof are consistent by querying every candidates verification oracle and then recomputes the output to compare it with the image. The construction allows us to state the following theorem. restate

Theorem 1.3

Relaxed Combiners for Unbiasable VRF. Having relaxed the unique provability property of the candidates we consider combiner constructions for more advanced notions of VRFs. A VRF is fully unbiasable if no efficient adversary can craft a set of keys such that the output distribution distribution is skewed into any direction. We show that fully unbiasable VRF as defined in **??** admit non-uniform robust $(2;1)$ black-box combiners that base their property on collision resistant hash functions and 2-source randomness extractors. . Formally we formulate the following theorem. this is unclear

Theorem 1.4 restate

Relaxed Combiners for Aggregable VRF. Then we consider aggregable VRF we define in **??** which additionally allow the aggregation of many image-proof pairs into a single image-proof pair to realize efficient batch verification. Key-homomorphic, secret-key aggregable, VRF require that the evaluation algorithm forms a group homomorphism with respect to group structures on the secret-key space and the product of the output codomain and the proof space. Using efficient grouphomomorphism tests by [**?**] we can construct a $(n;1)$ robust combiner for statistically key-homomorphic VRF that works by publicly checking the homomorphism property before running the algorithm. restate

Theorem 1.5

For public-key aggregability

Relaxed Combiners for Constrainable VRF. Finally we consider constrainable VRF, defined in **??**. Constrainable VRF additionally allow the user to generate constrained keys, which only allow the evaluation on a constrained set of evaluation points.

Chapter 2

Notation and Preliminaries

2.1 Notation

For a positive integer n we define $[n] = \{1, \ldots, n\}$. Unless specified otherwise we use λ to denote the security parameter. A function $\varepsilon(n)$ is called negligible, if it vanishes faster than the inverse of any polynomial in n, formally defined in **??**. We say that an event E happens with overwhelming probability if $\Pr[E] \geq 1 - \varepsilon(\lambda)$ for a negligible function ε and we say that an E occurs with significant probability if $\Pr[E] \geq \frac{1}{p(\lambda)}$ for a polynomial p in λ. We use $\mathsf{poly}(n)$ to denote the set of functions polynomial in n and $\mathsf{negl}(n)$ to denote the set of functions negligible in n. For a set S we write $y \leftarrow_\$ S$ to indicate that the value of y is sampled uniformly at random from S, for a random variable X we write $y \leftarrow_\$ X$ to indicate that the value of y is sampled from the corresponding sample space with the underlying distribution, and for a randomized algorithm $\mathcal{A}$ we write $y \leftarrow_\$ \mathcal{A}(x)$ to assign to y the value of the output of $\mathcal{A}$ on x. We use $\leftarrow$ for simple assignment. We denote column vectors with lower-case bold letters (e.g. $\mathbf{x}$) . Unless specified otherwise we assume that any algorithm gets as input in the first argument the unary encoding of the security parameter 1^λ. Finally we use ppt. to abbreviate probabilistic polynomial time in context of turing machines and algorithms.

I needed this previously bu

Definition 2.1 (Polynomial Functions) *We say that a function $f : \mathbb{N} \to \mathbb{R}$ is polynomial in its argument λ if*

$$\exists c, d \in \mathbb{N}, \forall \lambda \in \mathbb{N}, f(\lambda) \leq c\lambda^d$$

Definition 2.2 (Negligible Functions) *We say that a function $f : \mathbb{N} \to \mathbb{R}$ is negligible in its argument λ if*

$$\forall c \in \mathbb{N}^+, \exists n_0 \in \mathbb{N}, \forall \lambda > n_0, |f(\lambda)| < \frac{1}{x^c}$$

2.2 Verifiable Functions

A verifiable function (VF) VF is a cryptographic primitive that implements a key-generation algorithm $\mathsf{Gen}_{\mathsf{VF}}$ to sample a key-pair $(\mathsf{sk}, \mathsf{pk})$ containing a secret and a public key, a deterministic evaluation algorithm $\mathsf{Eval}_{\mathsf{VF}}$ that allows the holder of a secret key sk to make evaluations at any points $x \in \{0,1\}^\lambda$ to obtain an image $y \in \mathcal{Y}$ and a corresponding proof $\pi \in \mathcal{P}$, and a verification algorithm that allows a different party with access to the corresponding public key pk to verify that y was correctly computed from x if given access to the proof π. The definition takes inspiration of the definition of a verifiable random function (VRF) as defined by [**?**], to reduce redundand when considering VRFs with additional or relaxed properties we define the properties seperately on top of a verifiable function (VF).

Definition 2.3 (Verifiable Function) *A verifiable function (VF)* VF *is a triple of efficient (ppt.) algorithms* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *with the syntax*

this is less elegant now

- $\mathsf{Gen}_{\mathsf{VF}}$*: The key-generation algorithm takes as input the unary encoding of the security parameter and produces a pair* $(\mathsf{sk}, \mathsf{pk}) \in \mathcal{SK} \times \mathcal{PK}$ *containing a secret and a public key.*
- $\mathsf{Eval}_{\mathsf{VF}}$*: The evaluation algorithm takes as input a tuple* (sk, x) *containing a secret key* $\mathsf{sk} \in \mathcal{SK}$ *and an evaluation point* $x \in \{0,1\}^\lambda$ *and produces an tuple* (y, π) *containing an image* $y \in \mathcal{Y}$ *and a corresponding proof* $\pi \in \mathcal{P}$*.*
- $\mathsf{Vfy}_{\mathsf{VF}}$*: The verification algorithm takes as input a tuple* (pk, x, y, π) *containing a public key* $\mathsf{pk} \in \mathcal{PK}$*, an evaluation point* $x \in \{0,1\}^\lambda$*, an image* $y \in \mathcal{Y}$ *and a proof* $\pi \in \mathcal{P}$ *and returns a bit* b *denoting acceptance or rejection.*

A VF can assume a range of different attributes. Based on the satisfied properties we arrive at specific constructions such as verifiable random functions (VRFs) or verifiable unpredictable functions (VUFs). We first describe the correctness property of a VF. Informally a VF is correct if for any key-pair produced honestly from the key-generation algorithm and any point $x \in \{0,1\}$ the output (y, π) of the evaluation alogrithm is accepted by the verification algorithm.

Definition 2.4 (ε-Correctness) *A verifiable function* $\mathsf{VF} = (\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is ε-correct, if for all security parameters $\lambda \in \mathbb{N}^+$ we have that*

$$\Pr\left[\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y, \pi) = 1 : \begin{array}{r}(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^\lambda) \\ x \leftarrow_{\$} \{0,1\}^\lambda \\ (y, \pi) \leftarrow_{\$} \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)\end{array}\right] \geq \varepsilon(\lambda)$$

An ε-correct VF is defined to be perfectly correct, or just correct, if $\varepsilon = 1$, and it is defined to be statistically correct if $\varepsilon = 1 - \mu$ for a negligible function μ.

While correctness requires all honestly generated images to be accepted by the verification algorithm we additionally demand that the verification algorithm rejects all maliciously generated images. We define the property of unique provability to be the property that requires that for every evaluation point $x \in \{0,1\}^\lambda$, and every public-key $\mathsf{pk} \in \mathcal{PK}$ only one image $y \in \mathcal{Y}$ is accepted by the verification algorithm.

Definition 2.5 (Unique Provability) *A verifiable function* VF*, with algorithms* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$*, defined to be uniquely provable if for all evaluation points* $x \in \{0,1\}^\lambda$*, a public-keys* $\mathsf{pk} \in \mathcal{PK}$ *(even maliciously crafted ones) and any two images-proof pairs* $(y_0, \pi_0), (y_1, \pi_1) \in \mathcal{Y} \times \mathcal{PK}$*:*

$$\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y_0, \pi_0) = 1 \land \mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y_0, \pi_0) = 1 \implies y_0 = y_1$$

We define a secure VF to be a VF that satisfies both correctness and unique provability. We define the computational security properties of pseudorandomness, weak pseudorandomness and unpredictability.

maybe separate correct and secure here or maybe omit it fully

Definition 2.6 ((t,ε)-Pseudorandomness) *A VF* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is defined to be* (t,ε)*-pseudorandom if for all security parameters* $\lambda \in \mathbb{N}$*, for all stateful adversaries* $\mathcal{A}$*, running in time at most* $t(\lambda)$*, the advantage of* $\mathcal{A}$ *in the prf security experiment outlined in* **??** *is at most* $\varepsilon(\lambda)$*, i.e. if*

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a VF is pseudorandom, if it is (t,ε)*-pseudorandom with* ε *being a negligible function and* t *being a polynomial, i.e.* $\varepsilon \in \mathsf{negl}(\lambda), t \in \mathsf{poly}(\lambda)$*.*

Definition 2.7 ((t,ε)-weakly Pseudorandomness) *A VF* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is defined to be* (t,ε)*-weakly pseudorandom if for all security parameters* $\lambda \in \mathbb{N}$*, for all stateful adversaries* $\mathcal{A}$*, running in time at most* $t(\lambda)$*, the advantage of* $\mathcal{A}$ *in the wprf security experiment outlined in* **??** *is at most* $\varepsilon(\lambda)$*, i.e. if*

$$\mathbf{Adv}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a VF is weakly pseudorandom, if it is (t,ε)*-weakly pseudorandom with* ε *being a negligible function and* t *being a polynomial, i.e.* $\varepsilon \in \mathsf{negl}(\lambda), t \in \mathsf{poly}(\lambda)$*.*

Definition 2.8 ((ε,t)-Unpredictability) *A VF* $(\mathsf{Gen}_{\mathsf{VF}}, \mathsf{Eval}_{\mathsf{VF}}, \mathsf{Vfy}_{\mathsf{VF}})$ *is defined to be* (t,ε)*-unpredictable if for all security parameters* $\lambda \in \mathbb{N}$*, for all stateful adversaries* $\mathcal{A}$*, running in time at most* $t(\lambda)$*, the advantage of* $\mathcal{A}$ *in the up security experiment outlined in* **??** *is at most* $\varepsilon(\lambda)$*, i.e. if*

Experiment $\mathbf{Exp}^{\text{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

$b \leftarrow_{\$} \{0,1\}$
$(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
$S \leftarrow \varnothing$
$(x^*, \sigma) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{eval}}}(1^\lambda, \mathsf{pk})$
$(y_0, \pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
$y_1 \leftarrow_{\$} \mathcal{Y}$
$b' \leftarrow_{\$} \mathcal{A}(1^\lambda, \sigma, y_b)$
return $b = b' \wedge x^* \notin S$

Experiment $\mathbf{Exp}^{\text{wprf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

$b \leftarrow_{\$} \{0,1\}$
$(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
$S \leftarrow \varnothing$
$x^* \leftarrow_{\$} \{0,1\}^\lambda$
$(y_0, \pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
$y_1 \leftarrow_{\$} \mathcal{Y}$
$b' \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{eval}}}(1^\lambda, \mathsf{pk}, x^*, y_b)$
return $b = b' \wedge x^* \notin S$

Experiment $\mathbf{Exp}^{\text{up}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

$(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^\lambda)$
$S \leftarrow \varnothing$
$(x^*, y^*) \leftarrow_{\$} \mathcal{A}^{\mathcal{O}_{\text{eval}}}(1^\lambda, \mathsf{pk})$
$(y, \pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
return $y^* = y \wedge x^* \notin S$

Oracle $\mathcal{O}_{\text{eval}}(x)$

$S \leftarrow S \cup \{x\}$
$(y, \pi) \leftarrow_{\$} \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x)$
return y

Figure 2.1: Security game for pseudorandomness, weak pseudorandomness and unpredictability.

$$\mathbf{Adv}^{\text{up}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\text{up}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

We say that a VF is unpredictable, if it is $(\mathfrak{t}, \varepsilon)$*-unpredictable with* ε *being a negligible function and* $\mathfrak{t}$ *being a polynomial, i.e.* $\varepsilon \in \mathsf{negl}(\lambda), \mathfrak{t} \in \mathsf{poly}(\lambda)$.

We define a verifiable random function (VRF) to be a secure pseudorandom VF, a verifiable unpredictable function (VUF) to be a secure unpredictable VF and a weaker verifiable random function (wVRF) to be a secure weakly pseudorandom VF. We formally define the primitives VRF, VUF and wVRF in the following definitions.

Definition 2.9 (Verifiable Random Functions) *A verifiable random function* $(\mathsf{Gen}_{\mathsf{VRF}}, \mathsf{Eval}_{\mathsf{VRF}}, \mathsf{Vfy}_{\mathsf{VRF}})$ *is a VF that satisfies perfect correctness, unique provability and pseudorandomness.*

Definition 2.10 (Verifiable Unpredictable Functions) *A verifiable random function* $(\mathsf{Gen}_{\mathsf{VRF}}, \mathsf{Eval}_{\mathsf{VRF}}, \mathsf{Vfy}_{\mathsf{VRF}})$ *is a verifiable function that satisfies perfect correctness, unique provability and unpredictability.*

Definition 2.11 (Weak Verifiable Random Functions) *A verifiable random function* $(\mathsf{Gen}_{\mathsf{VRF}}, \mathsf{Eval}_{\mathsf{VRF}}, \mathsf{Vfy}_{\mathsf{VRF}})$ *is a verifiable function that satisfies perfect correctness, unique provability and weak pseudorandomness.*

Unbiasability, following the definition of **[?]**, is a property for VRFs that prevents adversaries from skewing the output distribution of the evaluation function on malleable keys. To formally definine unbiasability we require the notion of monotone predicates.

Definition 2.12 (Monotone Predicates [?]) *A probabilistic predicate* ρ *on the domain* $(\mathcal{Y} \cup \{\perp\})^*$ *is monotone, iff for all* $y \in \mathcal{Y}^*$*, all index sets* $I \subseteq [|y|]$ *and every bitstring of randomness* $r \in \{0,1\}^\lambda$ *we have*

$$\Pr[\rho(\tilde{y}; r) = 1] \leq \Pr[\rho(y; r) = 1]$$

where $\tilde{y}_i = y_i$ *if* $i \notin I$ *and* $\tilde{y}_i = \perp$ *otherwise.*

Definition 2.13 (Unbiasability [?]) *A VRF* VRF *is (fully)* $(\mathfrak{t}, \varepsilon, \mathfrak{a}, \mathfrak{b})$*-unbiasable if for every stateful adversary* $\mathcal{A}$ *running in time at most* $\mathfrak{t}$ *its advantage in the unbiasability security experiment in* **??** *is bounded by*

$$\mathbf{Adv}^{\mathrm{fub}}_{\mathsf{VRF}}(\lambda, \mathcal{A}) := 2 \cdot \Pr\left[\mathbf{Exp}^{\mathrm{fub}}_{\mathsf{VRF}}(\lambda, \mathfrak{a}, \mathfrak{b}, \mathcal{A}) = 1\right] - 1 \leq \varepsilon(\lambda)$$

We say that a VRF is unbiasable if it is $(\mathfrak{t}, \varepsilon, \mathfrak{a}, \mathfrak{b})$*-unbiasable and* ε *is a negligible function in the security parameter* λ*, i.e.* $\varepsilon \in \mathsf{negl}(\lambda)$.

Key-homomorphism of VF requires that the secret-key space and the product space of the image and proof space admit group structures $(\mathcal{SK}, \odot)$ and $(\mathcal{Y} \times \mathcal{P}, \otimes)$ respectively. Intuitively a VF is key-homomorphic if for every evaluation point $x \in \{0,1\}^\lambda$ the evaluation algorithm on x implements a grouphomomorphism in the secret key.

Definition 2.14 (Key-Homomorphism) *A VF* VF *is key-homomorphic if for every evaluation point* $x \in \{0,1\}^\lambda$*, secret key* $\mathsf{sk}_1, \mathsf{sk}_2 \in \mathcal{SK}$ *we have*

$$\mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}_1 \odot \mathsf{sk}_2, x) = \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}_1, x) \otimes \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}_2, x)$$

We also refer to key-homomorphism as secret-key aggregability. To define public-key aggregability we first define the cryptoraphic primitive of an aggregable verifiable function (AggVF) that additionally implements algorithms to combine image-proof pairs into a single image-proof pair aggregate.

think I should abandon the subsetting of the primitive and introduce . notation. Also use subexp **P** for primitive in combiner

Experiment $\mathbf{Exp}^{\mathrm{fub}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

$b \leftarrow_{\$} \{0,1\}$
$(\rho, \mathsf{pk}_1, \ldots, \mathsf{pk}_{\mathfrak{a}(\lambda)}) \leftarrow_{\$} \mathcal{A}(1^\lambda)$
require ρ is monotone
$x_1, \ldots, x_{\mathfrak{b}(\lambda)} \leftarrow_{\$} \{0,1\}^\lambda$
$(y^0_{1,1}, \pi^0_{1,1}), \ldots, (y^0_{\mathfrak{a}(\lambda),\mathfrak{b}(\lambda)}, \pi^0_{\mathfrak{a}(\lambda),\mathfrak{b}(\lambda)}) \leftarrow_{\$} \mathcal{A}(x_1, \ldots, x_{\mathfrak{b}(\lambda)})$
for $i \in [\mathfrak{a}(\lambda)], j \in [\mathfrak{b}(\lambda)]$ **do**
  $\beta_{i,j} \leftarrow \mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_i, x_j, y^0_{i,j}, \pi^0_{i,j})$
  if $\beta_{i,j} = 1$ **then**
    $z_{i,j} \leftarrow y^b_{i,j}$
  else $z_{i,j} \leftarrow \perp$
$r \leftarrow \{0,1\}^\lambda$
return $\rho(z_{1,1}, \ldots, z_{\mathfrak{a}(\lambda),\mathfrak{b}(\lambda)}, r) \neq b$

Figure 2.2: Full unbiasability security experiment.

2.3 Robust Combiners

Given a cryptographic primitive P we define a robust $(n;k)$ P-combiner

Definition 2.15 (Robust $(n;k)$ P-combiner) *Given a cryptographic primitive* P *describing p algorithms* $(\mathsf{alg}_\mathsf{P}{}^1, \ldots, \mathsf{alg}_\mathsf{P}{}^2)$*, a robust* $(n;k)$ P*-combiner is a p-tuple of algorithms that take as input the description of n Turing machines* $\mathcal{O}_1, \ldots, \mathcal{O}_n$*, the candidate schemes*

$$\Pi_\mathbf{P}(\mathcal{O}_1, \ldots, \mathcal{O}_n) = \left(\Pi_\mathbf{P}(\mathcal{O}_1, \ldots, \mathcal{O}_n).\mathsf{alg}_\mathsf{P}{}^1, \ldots, \Pi_\mathbf{P}(\mathcal{O}_1, \ldots, \mathcal{O}_n).\mathsf{alg}_\mathsf{P}{}^n\right)$$

that run in time polynomial in n, and the size of their input, such that the combiner securely implements the primitive P *if at least k candidates schemes securely implement the primitive* P*.*

We write robust combiner for robust $(n;k)$ P-combiner whenever n, k and P are clear from the context. A robust combiner gets the candidate schemes as a description of their turing machine and can therefore be adaptive in the code of its candidates. Note that this definition allows the robust combiner to ignores its candidate schemes and implement a secure instance of the underlying primitive indepdently of its candidates. One can therefore always construct a robust combiner for primitives that admit secure instances, by contraposition this also means that any result that rules out the existence of a robust combiner for a specific primitive P also rules out the existence

of the primitive itself. Intuitively we would like for the combiner to derive its security from the security of its secure candidate schemes, therefore we define robust black-box combiner. A robust black-box combiner is a robust combiner that, instead of getting access to the description of the Turing machines of the candidate schemes, only gets oracle access to the underlying candidate schemes. We define the notion of a robust black-box combiner closely following the definition by [**?**]. Note that we explicitly require a robust black-box combiner to derive its security from its candidates, for that reason we require the security to hold relative to a PSPACE oracle PSPACE.

Very unhappy with this definition

Definition 2.16 ($(n;k)$-Robust Black-Box P-Combiner) *Given a cryptographic primitive* P *describing p algorithm* $(\mathsf{alg_P}^1, \ldots, \mathsf{alg_P}^p)$*, a robust* $(n;k)$ *black-box* P*-combiner is a p-tuple of oracle Turing machines with access to n oracles, the candidate schemes* $\mathcal{O}_1, \ldots, \mathcal{O}_n$

$$\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n} = \left(\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{alg_P}^1, \ldots, \Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}.\mathsf{alg_P}^n\right)$$

A $(n;k)$*-robust combiner* $\Pi_{\mathbf{P}}$ *for primitive P is defined to be black-box, denoted by* Π_{P}*, if additionally it satisfies that if there exists an adversary* $\mathcal{A}$ *(possibly computationally unbounded) that breaks the security of* $\Pi_{\mathsf{P}}^{\mathcal{O}_1,\ldots,\mathcal{O}_n}$*, then there exist at black-box reductions with oracle access to* $\mathcal{A}$ *for at least* $n - k + 1$ *candidates.*

We remark that compositite security definitions that rely on multiple properties are broken if there exists an adversary that breaks a security property with significant advantage. Note that if the security would not hold relative to a PSPACE oracle PSPACE then we would find that a secure implementation of the primitive would indeed implement a secure robust black-box combiner since there exists no adversary breaking the security properties of P.

Chapter 3

Results

3.1 Robust Black-Box DAG Combiner

Intuitively, we imagine a robust black-box combiner to operate by orchestrating a sequence of queries to its candidate schemes with computations that we model with probabilistic turing machines in between. To make the combiner structure more explicit we will model it with a vertex-colored directed acyclic graph (DAG) where the nodes correspond to an oracle call to a candidate scheme specified by the vertex-coloring with inputs computed from the outputs of the oracle calls of the predecessors using a ppt algorithm. Formally we introduce the notion of a robust black-box DAG combiner:

Definition 3.1 (Robust $(n;k)$ Black-Box DAG P-Combiner) *Given a cryptographic primitive* P *implementing a set of algorithms indexed by* $j \in [|\mathsf{P}|]$. *A robust* $(n;k)$ *black-box DAG* P*-combiner is a* $|\mathsf{P}|$*-tuple*

$$\left(\mathfrak{G}_j = \left(\mathcal{G}_j, c_j, s_j, t_j, \{f_{j,v}\}_{v \in V(\mathcal{G}_j)}\right)\right)_{j \in [|\mathsf{P}|]}$$

where $\mathcal{G}_j$ *is a directed acyclic graph with vertex set* $V(\mathcal{G}_j)$, $c_j : V(\mathcal{G}_j) \setminus \{s_j, t_s\} \to [n] \times [|\mathsf{P}|]$ *is a vertex-coloring with colors* (a, b) *indicating that the corresponding query is made to algorithm* a *of candidate oracle* b, *source and sink* $s_j, t_s \in V(\mathcal{G}_j)$ *satisfying* $\mathcal{N}^-(s_j) = \mathcal{N}^+(t_j) = \varnothing$ *and for simplicity the only vertices satisfying those properties and finally* $\{f_{j,v}\}_{v \in V(\mathcal{G}_j)}$ *are ppt. Turing machines. satisfying that the following tuple of oracle Turing machines is a robust* $(n;k)$ *black-box combiner.*

$$\widetilde{\Pi^{\mathcal{O}_1,\ldots,\mathcal{O}_n}_{\mathsf{P}}} = \left(\widetilde{\Pi^{\mathcal{O}_1,\ldots,\mathcal{O}_n}_{\mathsf{P}}}.[\![\mathfrak{G}_1]\!], \ldots, \widetilde{\Pi^{\mathcal{O}_1,\ldots,\mathcal{O}_n}_{\mathsf{P}}}.[\![\mathfrak{G}_{|\mathsf{P}|}]\!]\right)$$

Where $[\![\mathfrak{G}_i]\!]$ *is the interpretation according to* **??**.

Algorithm $[\![\mathcal{G}, c, s, t, \{f_v\}_{v\in V(\mathcal{G})}; \rho]\!](\mathbf{x})$

1 : $\mathbf{y}_s \leftarrow \mathbf{x}$
2 : **for** $v \in \tau(\mathcal{G}, s) \setminus \{s, t\}$ **do**
3 : $\quad \mathbf{x}_v \leftarrow f_v(\mathbf{x}, \{(\mathbf{y}_w, w)\}_{w\in\mathcal{N}^-(v)}; \rho)$
4 : $\quad \mathbf{y}_v \leftarrow \mathcal{O}_{c(v)}(\mathbf{x}_v)$
5 : **return** $f_t(\mathbf{x}, \{\mathbf{y}_w\}_{w\in\mathcal{N}^-(t)})$

Algorithm $\tau(\mathcal{G}, s)$

1 : // Topological ordering TODO

Figure 3.1: Interpretation of $\left(\mathcal{G}, c, s, t, \{f_v\}_{v\in V(\mathcal{G})}\right)$ into an algorithm. τ returns the vertices in $\mathcal{G}$ in topological order starting from the source s.

Example. We show the robust $(2;1)$ black-box DAG combiner for verifiable random functions with guaranteed unique provability in **??** by drawing the graphs corresponding to each algorithm and labling each vertex with the corresponding oracle queried and the explicit description Turing machine modeling the preceeding computation. For clarity we will use the name of the algorithms rather than an indexing and we label the vertices with the corresponding oracle calls direclty.

$$s_{\mathsf{Gen}_{\mathsf{VRF}}}$$

$$\mathcal{O}_1.\mathsf{Gen}_{\mathsf{VRF}} \qquad \mathcal{O}_2.\mathsf{Gen}_{\mathsf{VRF}}$$

$$t_{\mathsf{Gen}_{\mathsf{VRF}}}$$

Figure 3.2: Drawing of the graphs $\mathcal{G}$

We will refer to a robust black-box DAG combiner as a DAG combiner. According to the definition **??** every DAG combiner is necessarily also a robust black-box combiner. In the following we show that the converse also holds, that is, that the class of DAG combiners and robust black-box combiners are computationally equivalent for any underlying primitive.

Theorem 3.2 (Generality of Robust $(n;k)$ Black-Box DAG Combiners) *For every robust $(n;k)$ black-box combiner there exists a computationally equivalent robust $(n;k)$ black-box DAG combiner.*

Informally this equality holds as every robust black-box combiner consists of algorithms that performs computations interleaved with oracle queries to their candidate schemes. We can unroll this computation create a node for each oracle query performed during the computation to arrive at the corresponding vertex-colored DAG, the computations performed in-between oracle queries can be represented with a ppt. Turing machine $f_{j,i}$.

Proof (??) Let P be a cryptographic primitive Π_{P} with defining algorithms $(p_1, \ldots, p_{|\mathsf{P}|})$, et Π_{P} be a robust $(n;k)$ black-box P-combiner according to **??**, and let $j \in [|\mathsf{P}|]$ be arbitrary. By **??** there exists a polynomial $T \in \mathsf{poly}(n, \lambda, |x|)$ where x is the input to p_j. We construct a directed acyclic graph $\mathcal{G}_j$

$$\mathcal{G}_j = ([n] \times [|\mathsf{P}|] \times [T] \cup \{s, t\}, \{\, ((a_1, b_1, t_1), (a_2, b_2, t_2)) : t_1 < t_2 \,\})$$

We define the coloring by $c(a, b, t) = (a, b)$. We define the Turing machines $\{f_{j,v}\}_{v \in V(\mathcal{G}_j)}$ iteratively starting from the beginning of the computation s, we define $f_{j,v_{j,a,b,1}}$ to be the ppt Turing machine that simulates the $\Pi_{\mathsf{P}}.p_j$ until the first call to its oracle. If that oracle happens to be (a, b) then we end, otherwise write 0 onto the tape. In every subsequent level $f_{j,a,b,t+1}$ simulates the combiner on and answers the queries to the respective oracle with the output of that query ($y_{j,a,b,k}$ for $k \in [t]$) by progressively simulating all previous Turing machines $f_{j,a,b,k}$ for $k \in [t]$. By checking whether after the computation we find a 0 on the tape we can trace the path of queried oracles. We can now continue the simulation of the combiner since we know exactly the state of the tape. The combiner is necessarily efficient since for every the number of vertices in the graph is bounded from above by $nT(n, \lambda, |x|)|\mathsf{P}|$ where $T \in \mathsf{poly}(m, \lambda, |x|)$, we only make exactly one query to any oracle per vertex (which also puts a polynomial bound on the number of oracle calls), the topological ordering can be computed in polynomial time. We are only left to show that the Turing machines $\{f_{j,v}\}_{v \in V(\mathcal{G}_j)}$ are indeed ppt. To this end note that $f_{j,v}$ simulates the black-box combiner Turing machine, which is assumed to be ppt, at most once where it can answer any oracle query by looking up the corresponding output of the layer in time polynomial in $n, |\mathsf{P}|$ and λ. □

3.2 Impossibility of Robust Black-Box VRF-Combiner

We consider the possibility of combining VRFs in a black-box fashion. As previously mentioned we present an impossibility result for general robust $(n; n-1)$ black-box VRF-combiners. This means that given n VF oracles there is no way to combine them in a construction such that the construction is VRF secure if at least $n-1$ candidates are secure. For simplicity we assume that the secret key spaces of the candidates are of the form $\mathcal{SK} = \{0,1\}^{s(\lambda)}$ for some $s \in \mathsf{poly}(\lambda)$.

Theorem 3.3 *There exists no robust* $(n; n-1)$ *black-box* VRF*-combiner.*

Intuition. We need to show that for every $(n; n-1)$ black-box VRF-combiner Π_{VRF} there exists a tuple $\mathbf{O} = (\mathcal{O}_1, \ldots, \mathcal{O}_n)$ of VF oracles of which at least $n-1$ implement a secure VRF, the combiner $\Pi^{\mathbf{O}}_{\mathsf{VRF}}$ doesn't implement a secure VRF. By equivalence of robust black-box combiner and robust black-box DAG combiners in **??** we instead show the statement for robust black-box DAG combiners. For that we fix an arbitrary robust black-box DAG combiner Π_{VRF} with the DAGs $G_{gen}, G_{eval}, G_{vfy}$ with sources $s_{gen}, s_{eval}, s_{vfy}$, sinks $t_{gen}, t_{eval}, t_{vfy}$, colorings $c_{gen}, c_{eval}, c_{vfy}$ and ppt. algorithms $\{f_v^{gen}\}_{v \in V(G_{gen})}$, $\{f_v^{eval}\}_{v \in V(G_{eval})}$ and $\{f_v^{vfy}\}_{v \in V(G_{vfy})}$ for the algorithms $\mathsf{Gen}_{\mathsf{VRF}}, \mathsf{Eval}_{\mathsf{VRF}}$ and $\mathsf{Vfy}_{\mathsf{VRF}}$ respectively. We fix a topological ordering of vertices $v_1, \ldots, v_q$ in G_{eval} and iterate over them. Since we assume verification and evaluation algorithms to be deterministic the the randomness from queries to the key-generation algorithm cannot influence the evaluation or verification output, therefore those queries can replaced with a hard-coded key. Additionally since all oracle queries to eval and verification are determiniistic we assume that for every vertex the query input for each oracle is unique (since we can just connect an edge to the first query, in topolofical ordering, and use this instead of requerying). We finally make the assumption that $\mathcal{SK} = \{0,1\}^{s(\lambda)}$ for $s \in \mathsf{poly}(\lambda)$.

Let $v_1, \ldots, v_n$ be a topological ordering of the vertices in G_{eval}. We iterate over the vertices in the order $v_1, \ldots, v_n$ and distinguish the following cases: Either for all uniquely provable and perfectly correct candidate oracles $\mathcal{O}_1, \ldots, \mathcal{O}_n$ and secret keys sk, and evaluation points x, the output of the query made corresponding to v_i does not influence the output at all, that is if we change the output of the query associated to v_1 the output y of the combiner remains the same. We call such vertices *irrelevant*. Irrelevant vertices can be simulated by an adversary by using any arbitrary value.

Since we know that all prior vertices are irrelevant we know that the function associated with the vertex v_i, $f_{v_i}^{eval}(\mathsf{sk}, x, o_1, \ldots, o_{d^+(v_i)})$, can be replaced by a function $\tilde{f}_{v_i}^{eval}(\mathsf{sk}, x) = f_{v_i}^{eval}(\mathsf{sk}, x, r_1, \ldots, r_{d^+(v_i)})$ where the values of r are obtained by arbitrary values in the support of the corresponding oracles.

need to add argument why replacing the output of irrelevant vertices with arbitrary evaluations preserves unique provability

Assume for all uniquely provable and perfectly correct candidates $\mathbf{O}$ we have that replacing the true sk in the execution of the evaluation algorithm with another $\widehat{\mathsf{sk}}, \widetilde{\mathsf{sk}} \leftarrow_{\$} \Pi_{\mathsf{VRF}}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$ the output of the combiner doesnt change, except with negligible probability, then this query is also simulatable as the adversary can just query the corresponding oracle on (sk', x) where $\mathsf{sk}' \leftarrow_{\$} \Pi^{\mathbf{O}}_{\mathsf{VRF}}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$ and with overwhelming probability the oracle produces the same output. We call those queries *public*. Public queries can also easily be simulated.

If we arrive at a non-irrelevant and non-public vertex v_j then we first claim that the verification algorihm necessarily needs to requery the exact same query. If that were not the case then we can construct VRF secure candidates (except for the oracle corresponding to this non-irrelevant and non-public vertex), and only let this oracle differ for that exact evaluation. Since at least $n-1$ candidates are VRF secure, the combiner also needs to be VRF secure, in particular uniquely provable and perfectly correct. By perfect correctness the two output however $\hat{y} \neq \tilde{y}$ both need to verify in their respective world. On the other hand in the other world they can not verify unless they break unique provability of the combiner. Since the two worlds can only be distinguished by this single oracle evaluation we conclude that the verification algorithm must rerun this query.

Let $T \subseteq V(G_{eval})$ be the set of vertices that are not irrelevant and not public. We now construct the following pseudorandomness adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$. $\mathcal{A}_1$ plays the experimentation phase of the prf security experiment, it computes $2s(\lambda)$ many input-evaluation pairs $\{(x_i, y_i, \pi_i)\}_{i \in 2s(\lambda)}$. Now it iterates over all evaluation pairs $i \in [2s(\lambda)]$ and iteratively filters the set of secret keys $\mathcal{SK}$ using its PSPACE oracle PSPACE. The filtering works as follows: For every secret key candidate $\mathsf{sk} \in \mathcal{SK}'$ simulate the computation of the evaluation algorithm, irrelevant and public vertices are easily simulated. For vertices in T follow the evaluation graph and once we would make a query we check whether the query q is in the transcript $\mathcal{T}_i$, if the immediate query doesn't occur in $\mathcal{T}_i$ the secret key sk can be thrown out. Otherwise if it occurs in $\mathcal{T}_i$ then we continue with that oracle response. If the final output y differs from the obtained y_i then we can also toss it outside, otherwise the secret key sk remains a valid candidate.

Note that due to correctness we never discard the correct secret key $\mathsf{sk}^* \in \mathcal{SK}$, if in every second filtration we drop at most half the secret keys from the previous candidates then with high probability, on a uniformly random value $x \leftarrow_{\$} \{0,1\}^*$ we win the pesudorandomness game. Otherwise if at least half of the previous candidates are discarded we end up with a polynomially bounded set of remaining secret keys $\mathcal{SK}^*$ from which we can guess the correct secret key with significant success probability and also win the pseudorandomness game.

Proof (??) Fix a robust $(n; n-1)$ black-box DAG VRF-combiner Π_{VRF} and consider the directed acyclic graph $\mathcal{G}_{eval}$ corresponding to the evaluation algorithm (see **??**). Additionally fix a secret key $\mathsf{sk} \in \mathcal{SK}$ and $x \in \{0,1\}^\lambda$. We describe the pseudorandomness adversary $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ as follows: $\mathcal{A}_1$ samples $2s(\lambda)$ many evaluation points $x_i \leftarrow_\$ \{0,1\}^\lambda$ uniformly at random and obtain the evaluation from the PRF oracle $\Pi^{\mathsf{O}}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, \cdot)$ to construct $\{(x_i, y_i, \pi_i)\}_{i \in [2s(\lambda)]}$. For each tuple (x_i, y_i, π_i) the adversary, given the public key pk computes a transcript of the combiners verification algorithm $\mathcal{T}_i$ which contains all oracle queries and responses made as well as the computed functions. Now additionally he computes *default* values $z^{eval}_{i,j} := \mathcal{O}_i.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}^*_i, x_j)$ for all $i \in [n]$ and $j \in [2s(\lambda)]$ and a $\mathsf{sk}^* \leftarrow_\$ \mathcal{O}_i.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$. For each $i = 1, \ldots, 2s(\lambda)$ the adversary $\mathcal{A}_1$ computes $\mathcal{SK}_i$ which is defined as:

$$\mathcal{SK}_i := \left\{\mathsf{sk} \in \mathcal{SK}_{i-1} : \tilde{f}(\mathsf{sk}, x_i, \mathcal{T}_i) = (y_i, \pi_i)\right\} \text{ where } \mathcal{SK}_0 := \mathcal{SK}$$

Where $\tilde{f}(\mathsf{sk}, x_i)$ runs the evaluation algorithm of the combiner and for every query to an oracle $\mathcal{O}_i$ we simulate the response by first checking the transcript $\mathcal{T}_i$ whether an identical query has been made to use, and otherwise use the default value $z^{eval}_{i,j}$. Since $\tilde{f}(\mathsf{sk}, x_i, \mathcal{T}_i)$ doesnt query any oracles, we can compute all of this us§ing the PSPACE complete oracle PSPACE. We now argue that following the topological ordering of the oracle queries (vertices in $\mathcal{G}^{eval}$), if a query is not exactly repeated in the combiners verification algorithm, and thus available in the transcript, the query itself is *irrelevant* in the sense that if replaced with $z^{eval}_{i,j}$ the evaluation of the combiner doesn't exist. We formally state this result in the following claim:

Claim 3.4 *For a fixed, hidden secret key* sk *and any arbitrary value of* $x \in \{0,1\}^\lambda$ *we have that if an oracle query* $\mathcal{O}_{i,j}(\widetilde{\mathsf{sk}}, \widetilde{x})$ *is done in the evaluation algorithm* $\Pi^{\mathsf{O}}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) = (y, \pi)$ *then the query is either exactly repeated in the evaluation of the combiners verification algorithm on the corresponding public key and* y, π*:* $\Pi^{\mathsf{O}}_{\mathsf{VRF}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)$ *or the value of* y *stays the same even if the output of* $\mathcal{O}_{i,j}(\widetilde{\mathsf{sk}}, \widetilde{x})$ *is replaced with any value* $z^{eval}_{i,j}$ *that is in the support of* $\mathcal{O}_{i,j}$*, that is if the query made to* $\mathcal{O}_{i,j}$ *is irrelevant.*

To prove **??** we assume towards a contradiction, that the query to $\mathcal{O}_{i,j}(\widetilde{\mathsf{sk}}, \widetilde{x})$ is neither repeated in $\mathcal{T}_i$, nor irrelevant to the output y. This means that there exist two possible outputs of oracle $\mathcal{O}_{i,j}(\widetilde{\mathsf{sk}}, \widetilde{x})$ that result in different evaluations $y^{(0)} \neq y^{(1)}$. This implies that the combiner either is not perfectly correct or uniquely provable and must either accept both $y^{(0)}, y^{(1)}$ or neither. By perfect correctness we have that $y^{(0)}$ must correctly verify in the real world and $y^{(1)}$ must correctly verify in a world where the oracle $\mathcal{O}_{i,j}$ output is remapped to the output resulting in the evaluation of $y^{(1)}$. Since however the oracle are never queried on the only different remapping of $\mathcal{O}_{i,j}$ both must be accepted in the real world, breaking unique provability.

Using the result of **??** we know that all queries we make that we cannot look up in the verifiers transcript are irrelevant in nature, that is we can replace a single one of them with a default value without changing the combiner evaluation. During evaluation we thus modify the first *unknown* oracle evaluation, i.e. the evaluation not in the transcript. During evaluation we replace it with a default value, once we reach the next unknown oracle evaluation.

For all oracles queries of a single oracle in $[n]$ we have that all queries made that are not repeated in the verification algorithm are as a group irrelevant for the output, that means that any of them can be overwitten / reprogrammed. This cannot be done over multiple oracles however

- Given a secret key sk of an oracle $\mathcal{O}_i$ it is still impossible for an efficient algorithm to find a secret key sk' such that $\mathsf{sk} \sim \mathsf{sk}'$ with non-negligible probability. That means that except with non-negligible probability we dont query an oracle on $\mathsf{sk}_0, \mathsf{sk}_1$ such $\mathsf{sk}_0 \sim \mathsf{sk}_1$ in one execution of the evaluation algorithm.
- We can modify the output of one candidate to break unique provability, perfect correctness and pseudorandomness. If we assume that the *individually irrelevant* are not *jointly irrelevant* then by defaulting the oracle output in topological ordering we necessarily arrive at an oracle query q for which defaulting changes the output. We **claim** that this output must therefore also be stored in the transcript of the verification, this is because we can simply rewire the previous oracles to assume the modified outputs to obtain $\mathcal{O}_1^*, \ldots, \mathcal{O}_n^*$ for which the evaluation of the verification algorithm doesnt change, BUT the evaluation of the combiner now would have a relevant and public query not being requeried by the verification.

  Note that for rewiring we need to make sure that the oracles $\mathcal{O}_1^*, \ldots, \mathcal{O}_n^*$ are still pseudorandom, uniquely provable and correct, i.e. implement secure VRFs. For all calls to eval we need to remap all calls to eval under *related* secret keys $\mathsf{sk}^* \sim \mathsf{sk}$ and all corresponding verification queries. For a verification query we need to rewire the corresponding evaluation call, the evaluation calls to all related secret keys $\mathsf{sk}^* \sim \mathsf{sk}$ and all corresponding verification queries.

  Now assume that we are at an evaluation query to rewire, the only way that the rewiring can impract the output of the combiners evaluation is if there is a *relevant* query to verify with an associated public key on the original oracle output y or if there is a relevant evaluation on a related secret key on that same input. If the latter were the case we could break related key recovery using the combiner. If the former is the case then the verification algorithm should come after the evaluation algorithm

since otherwise the evaluation combiner algorithm can predict the y. If it comes later then we can choose this oracle to be the malleable one, modify its output in the eval algorithm, forcing the verification to 0 at the same time TODO

- We claim that if the irrelevant queries are oracle-wide irrelevant, then they are overall also jointly irrelevant. Intuitively this must be the case since, if the oracles were not jointly irrelevant then one oracles irrelevant queries would only be irrelevant for a evaluations of another oracle, and relevant for another specific construction (if that were the case and the other construction would be a secure VRF oracle, then relevant the verification combiner needs to repeat those queries). Since neither that oracle, nor the combiner construction, however has knowledge of how that other oracle evaluates this can clearly not be possible. This would indeed only be possible if the combiner could limit those executions to not secure VRF oracles.

Now we just need to somehow get rid of the irrelevant oracle calls and replace them with a deterministic algorithm to do the key-filtering... How about a function $f(x, y)$ such that for a fixed $\hat{x}, \hat{y}$ we have $f(\hat{x}, y_0) = f(\hat{x}, y_1)$ and $f(x_0, \hat{y}) = f(x_1, \hat{y})$ for all y_0, y_1, x_0, x_1 but $f(x_0, y_0) \neq f(x_1, y_1)$.

One example is given by $f(x, y) = \max\{x, y\}$.

We now iterate over the oracles in topological ordering. For every query we precompute a default value which is equivalent to the corresponding oracle query on a random input, either (sk, x) or (pk, x, y, π) where sk are tested to not be *related*. Now we iterate over the oracle queries in topological ordering, for each query in the transcript we answer with the correct answer, for an answer in the cache we answer with the answer, for every query not in the transcript we default and store the mapping in the cache. Assume at step v_i the combiner output changes, as opposed to if we continue computation with the oracles, then we know that the query was not in the transcript (otherwise the execution would be the same). Since the query is not in the transcript and the query changes the combiner output this must mean that the combiner is necessarily not uniquely provable or perfectly correct.

We now aim to show that the combiner, however, must be uniquely provable and perfectly correct, since we can find VRF secure instanciation of the oracles that align with the evaluation transcript and the subsequent honest evaluation. Assume towards a contradiction that at least two candidates have no *VRF-secure extension*, that means that they cannot be extended from the existing mappings in a way such that at least $n - 1$ of them are VRF secure. That means that there must exist a conflict, we enumerate all possible conflicts at every time step i, assuming the prior modifications left the cnadidates all VRF secure (base case $i = 0$ where n are assumed to be secure)

- Assume we encounter an evaluation query with input (sk, x) and default it to (y, π). The only way that this would break unique provability or perfect correctness would be if there exists a related sk' which outputted $(y' \neq y, \pi)$. Assuming that related keys cannot efficiently be found this case only holds with negligible probability. Otherwise there exists a prior verification query which evaluated to $(\mathsf{pk}, x, y, \pi) \mapsto 0$ for pk associated with sk then we simply make this oracle the insecure one. If on the other hand there exists a verification query to $(\mathsf{pk},)$ problem
- Assume we encounter a verification query with input (pk, x, y, π) then this can only break a candidates unique provability or perfect correctness if there exists a verification on 1 (must be checked) on a related public key, we assume that also finding related public keys is hard. Or there must exist an evaluation on a related secret key sk, x that results in y, π with sk associated to pk. If that also was in the transcript that doesnt work, since we assume all candidates secure then. If the evaluation was defaulted the probability of that happening
- A query we defaulted was queried on eval for (sk', x) to (y', π') even though there is another query to (y, π) on (sk, x). In that case if $\mathsf{sk} \sim \mathsf{sk}'$ and $y' \neq y$ we have a contradiction to unique provability. However note that $\mathsf{sk} \sim \mathsf{sk}'$ requires the combiner to come up with two related secret keys, which we figure is impossible for efficient algorithms.
- A query we defaulted was queried on eval for (sk', x) to (y', π') even though there is another query to verify on $(\mathsf{pk}, x, y', \pi') = 0$ or on $(\mathsf{pk}, x, \tilde{y} \neq y', \tilde{\pi}) = 1$. We default all verification queries to 0 and exclude answeres on y' in simulation that we have seen. Either the evaluation is checked and requires evaluation to y' and we answer the verification with 0,

  Either the verification query evaluated to 1 by being in the transcript, then the output needed to be computed

Note that for all previous queries there is no conflict, either the queries are done to evaluation and there we **assume** that the evaluation algorithm cannot find two related queries, hance the default answeres just redefine outputs of the equivalence classes of related secret keys. Verification queries are answered with 1 on (pk, x, y, π) if and only if for all replays with a different choice of y of queries answered with y

If an oracle query cannot be found in the transcript or the cache we evaluate it We now want to show that if oracle-wide irrelevant queries can be remapped without changing the combiners evaluation, overall oracles the irrelevant queries are jointly irrelevant as well. We proceed with the following argument: By showing that we can remap those outputs in a way such that there exists an n-tuple of secure VRF oracles that matches this evaluation transcript in

topological ordering of the irrelevant vertices, then if we encounter a vertex which becomes relevant through that remapping this query must necessarily be repeated

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Assume, for a warm-up that $n = 2$. If we know that at least one candidate is VRF secure then we know that during the execution of evaluation of the combiner on (sk, x) any relevant query made by either oracle must be re-queried by the verification algorithm. If that were not the case then simply let this be the malleable oracle, then produce two outputs, which by perfect correctness both must verify, breaking unique provability.

Since this has to hold for both oracles, just initialize one of the two oracles with the malleable one, this execution becomes simulatable entirely. The other execution is given in parts by the transcript of the verification algorithm where the entries that are not contained in the transcript can simply be simulated with random evaluations.

We want to mention that instead of achieving weak pseudorandomness we can also allow unpredictability instead. As a corollary we can exclude the possibility of wVRF combiners in the sense of **??** and VUF combiners in the sense of **??**.

Corollary 3.5 (Impossibility of a Robust $(n; n-1)$ Black-Box VRF-combiner). *Given n secure VRF candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$ there exists no robust black-box combiner that implements a secure VRF scheme if even one candidate scheme is not uniquely provable.*

Corollary 3.6 (Impossibility of a Robust $(n; n-1)$ Black-Box VUF-combiner). *Given n secure VUF candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$ there exists no robust black-box combiner that implements a secure VUF scheme if even one candidate scheme is not uniquely provable.*

Having ruled out the possibility of combining VRFs, VUFs, wVRFs even for statistical notions of correctness and unique provability we need to make an even stronger assumption to allow for combiners to exist. We will assume that all candidates satisfy unique provability. While this might sound like a very strong assumption, which it is, we argue that it is not too hard of a restriction on modern schemes that pose potential candidate schemes for a combiner. This is because unique provability is an information theoretic notion which is typically mathematically proven for good constructions such as ECVRF or Lattice VRF.

However, assuming that all candidates satisfy unique provability combiner constructions become very simple. We present a construction very similar to the simple xor combiner for PRFs that works as follows. The key generation algorithm is returns a tuple of n secret keys and a tuple of n

what happens if the query input to one that was previously in the transcript changes as a result of the wrong but output-preserving guess of an irrelevant oracle. I think it will be the case that for any new transcript produced by guessing the output wont change, and a query done that would change the output must then be repeated in verification algorithm. That means we can just guess our way through.

public keys, where the i'th entry is an independently sampled key from candidate i, the evaluation algorithm evaluates x once for each candidate under the corresponding secret key and XORs the results together to obtain the image, additionally the proof is simply the transcript of the execution. The verification algorithm checks whether all oracle calls align by iterating over the queries made in the transcript and verifying each with a call to the corresponding verification oracle with the corresponding public key. Afterwards we recompute y by xoring the partial outputs of the candidates to verify that we obtain the same image. To prevent the combiner from being incorrect if a candidate it incorrect we execute a public checking routine before evaluating any algorithm. The public checking routine consists of sampling a polynomial number (in λ) of secret keys sk and evaluation points x, computing their evaluation (y, π) and then verifying with the corresponding public key pk for each oracle. If one oracle behaves incorrectly it is ignored during the execution of the combiner. Allowing the combiner to achieve statistcal correctness. Note that this combiner already works if only at least one candidate is secure. We state this result as a lemma below and show the construction of the combiner in **??**.

DOESNT WORK DUE TO UNIQUE PROVABILITY BREAK when collision is found but not remarked Here couldn't the evaluation simply involve a check to verify. If this check fails then a candidate fails, this is proven to the verification algorithm by providing an incorrect evaluation/verification in the proof. If the verifier can detect that fault then he accepts the skipping of the candidate, if he doesn't detect the fault then he cannot accept the skipping and rejects if that flag is set or if the other verification doesnt accept.

Lemma 3.7 *There exist a robust $(n;1)$ black-box combiner that combines n uniquely provable VF candidates, of which at least one implements a secure VRF to construct a VF that is statistically correct in the sense of* **??**, *uniquely provable in the sense of* **??** *and pseudorandom in the sense of* **??**.

Intuition. The proof of the security works in two stages. First we show that by doing the public correctness check we can detect incorrect candidates with overwhelming probability. Then we show that given any adversary $\mathcal{A}$ against the pseudorandomness of the combined scheme we can construct a reduction $\mathcal{R}$ against the pseudorandomness of any underlying candidate. The reduction works by simulating all other candidates faithfully, relying on the oracle in the prf security experiment to simulate the attacked candidate, and then use $\mathcal{A}$. Unique provability follows from unique provability of all underlying candidates.

Proof (??) should be easy

Algorithm $\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$

1 : $(\mathsf{sk}_1, \mathsf{pk}_1) \leftarrow_\$ \mathcal{O}_1.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$
2 : $(\mathsf{sk}_2, \mathsf{pk}_2) \leftarrow_\$ \mathcal{O}_2.\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda)$
3 : **return** $((\mathsf{sk}_1, \mathsf{sk}_2), (\mathsf{pk}_1, \mathsf{pk}_2))$

Algorithm $\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$

1 : $(\mathsf{sk}_1, \mathsf{sk}_2) \leftarrow \mathsf{sk}$
2 : $(y_1, \pi_1) \leftarrow \mathcal{O}_1.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_1, x)$
3 : $(y_2, \pi_2) \leftarrow \mathcal{O}_2.\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_2, x)$
4 : $y \leftarrow y_1 \oplus y_2$
5 : $\pi \leftarrow (y_1, \pi_1, y_2, \pi_2)$
6 : **return** (y, π)

Algorithm $\Pi^{\mathcal{O}_1,\mathcal{O}_2}_{\mathsf{VRF}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi)$

1 : $(\mathsf{pk}_1, \mathsf{pk}_2) \leftarrow \mathsf{pk}$
2 : $(y_1, \pi_1, y_2, \pi_2) \leftarrow \pi$
3 : **req** $y = y_1 \oplus y_2$
4 : **req** $\mathcal{O}_1.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_1, x, y_1, \pi_1) = 1$
5 : **req** $\mathcal{O}_2.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}_2, x, y_2, \pi_2) = 1$

Figure 3.3: Combiner for VRFs if all candidate schemes are uniquely provable.

Having shown that general notion of combiners on VRFs even with relaxed requirements on the security properties of the combiner are impossibile and the combiners on VRFs with guaranteed unique provability are easy we want to strengthen the result by considering combiners for VRFs (assuming all candidates are uniquely provable) with advanced additional properties, namely constrainability as in **??**, key-homomorphism as in **??**, aggregability as in **??** and finally unbiasability as in **??**.

For the notions of constrainability, key-homomorphism and aggregability this question is quckly answered as we we show how the above construction of extended PRF combiner can be minimally adapted to combine the notions. We will quickly explain how to extend the construction of the extended PRF combiner to capture those notions and then we will move on to the more interesting notion of unbiasability, for which the constructions become more complex.

We now consider additional security properties for VRFs, aggregatability that allows the aggregation of many secret keys or image-proof pairs to allow for efficient batch verification and unbiasability which prevents adversaries from skewing the output distribution of a VRF by maliciously selecting the keys.

Robust Black-Box Combiner for Unbiasable VRF. We present a robust black-box combiner for unbiasable VRF under the assumption that all candidate schemes are uniquely provable. We first note that there exists a compiler

to reduce unbiasable VRF to VRF under computational Diffie-Hellman (CDH) assumption [**?**], a combiner that works under the CDH assumption would be to ignore unbiasability of the candidates and combining normal VRFs and then introducing unbiasability afterwards . Instead of relying on an additional hardness assumption we prefer to use the unbiasability property of the secure candidates. We introduce a robust $(2;1)$ black-box combiner for unbiasable VRF based on collision resistant hash functions and randomness extractors that is heavily inspired by the construction of unbiasable VRF presented in [**?**]. The construction works by initially splitting the input $x \in \{0,1\}^\lambda$ into two parts $x = x_0 \| x_1$ then hashing $x_1 \| 0$ and $x_1 \| 1$ using a collision resistant hash function. Finally we evaluate each of the candidate on one point $H(x_1 \| 0)$ and $H(x_1 \| 1)$ before running a randomness extractor to extract randomness from x_0 and adding it to $y_0 \oplus y_1$. We formulate the result formally as a theorem.

is this even a robust black-box combiner since we cannot reduce unbiasability from the combiner to its candidates.

Theorem 3.8 (Robust $(2;1)$ Black-Box Combiner for Unbiasabile VRF) *There exists a uniform robust $(2;1)$ black-box combiner for unbiasable VRF if all candidate schemes are uniquely provable.*

Robust Black-Box Combiner for Aggregatable VRF. We first look at key-homomorphism combiners. Given a key-homomorphic function given an evaluation point $x \in \{0,1\}^\lambda$ we show how to publicly check for the homomorphism property of each candidate scheme. This is possible due to an algorithm by Blum [**?**] which introduces an efficient homomorphism check that runs in time independent of the group size. This means that given an evaluation point we can check whether the evaluation function is a group homomorphism with respect to the secret-key argument with high probability in time independent of the size of the secret-key space. We state this claim in form of a lemma and quickly elaborate on the homomorphism check, referring to the original result for the complete proof.

...

Lemma 3.9 (Robust $(2;1)$ Black-Box Combiner for Key-homomorphic VRF)

Next we consider aggregable VRFs. We show that indeed there exists a combiner

3.3 Discussion and Extensions

There are different ways to combine VRFs, for both unique provability and key-homomorphism they cannot be combined in a black-box fashion if we want to preserve properties such as unpredictability and therefore also randomness. On the other hand we manage to construct a robust black-box combiner that combines unbiasabile VRFs under guaranteed unique provability for all candidates. We find that the black-box combination of information

theoretic properties generally seems hard if other properties such as unpredictability need to be preserved. While the class of all robust black-box combiners initially seems to be very broad we show that we can capture all polynomially bounded combiners using the notion of a robust black-box DAG combiner, which allows a more systematic analysis of possible constructions by abstracting a graph structure in the computation executed by the implemented algorithms. We have not considered possible combiner constructions for stronger relaxations, such as computational unique provability or correctness with error, this is mainly because the resulting construction only offer weak guarantees that are insufficient for practical applications.

Chapter 4

Conclusion

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Appendix

Proof of ??

Proof It is sufficient to find a $(1;2)$-robust black-box combiner for VRF where all candidates are uniquely provable. Given a combiner we can then construct a $(1;n)$-robust black-box combiner for VRF by nesting the combiners iteratively. A construction for the combiner is given in **??**. The key generation works by querying both candidates on the given security parameter and grouping the secret and public keys respectively. Evaluation again queries both candidate schemes on the input with their respective secret key and returns the xor of the outputs while the the individual outputs and proofs consistute the new proof. The verification algorithm checks whether the image y is the result of xoring the individual outputs of the candidates and whether the individual images y_1, y_2 verify with respect to the verification key the proof and the pre-image. To ensure that the combiner is still efficient we sim

The combiner is clearly efficient since every algorithm consists of one call to the respective algorithm of each candidate scheme (whi)

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

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